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Chhanam Thorat 4963

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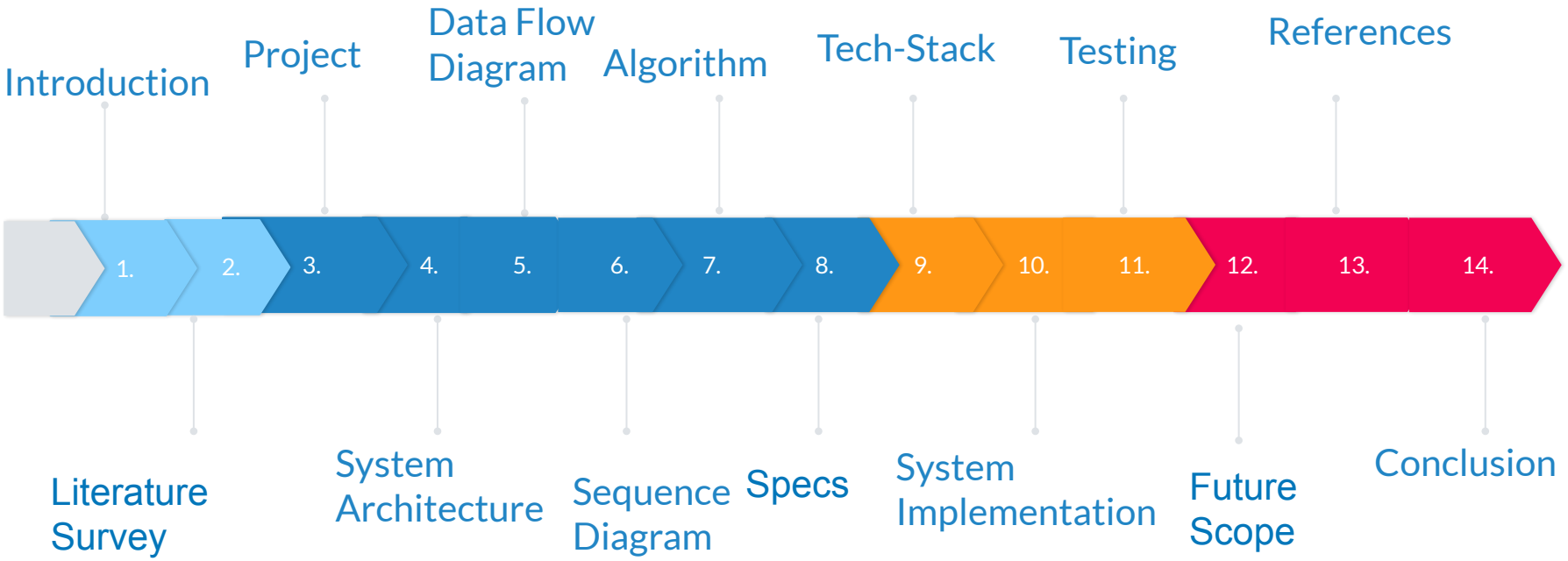
DATA EXTRACTION FROM DIGITIZED DOCUMENTS

Final Project Sem II ESE Presentation

Sponsored by: HEXANIKA

External Mentor: Mr. Satish Ranade

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Problem

Banks documents data entry & storage.

Manual data entry issues:

- Human error
- Maintenance of hard copies
- Cost Incurred to train employees

Solution

Perform Optical Character Recognition **(OCR)** on bank documents.

Store output machine readable text in CSV file for future use.

Saves efforts!

Scope

- The system is currently limited to document upload, extract data and storing in CSV file to download.
- User logs are stored in File System, no database.
- English characters and numerals only.
- Only LAN users can access.

Project Objectives



01.

To minimize the time & cost during the manual data entry.

02.

To achieve higher accuracy in text identification.

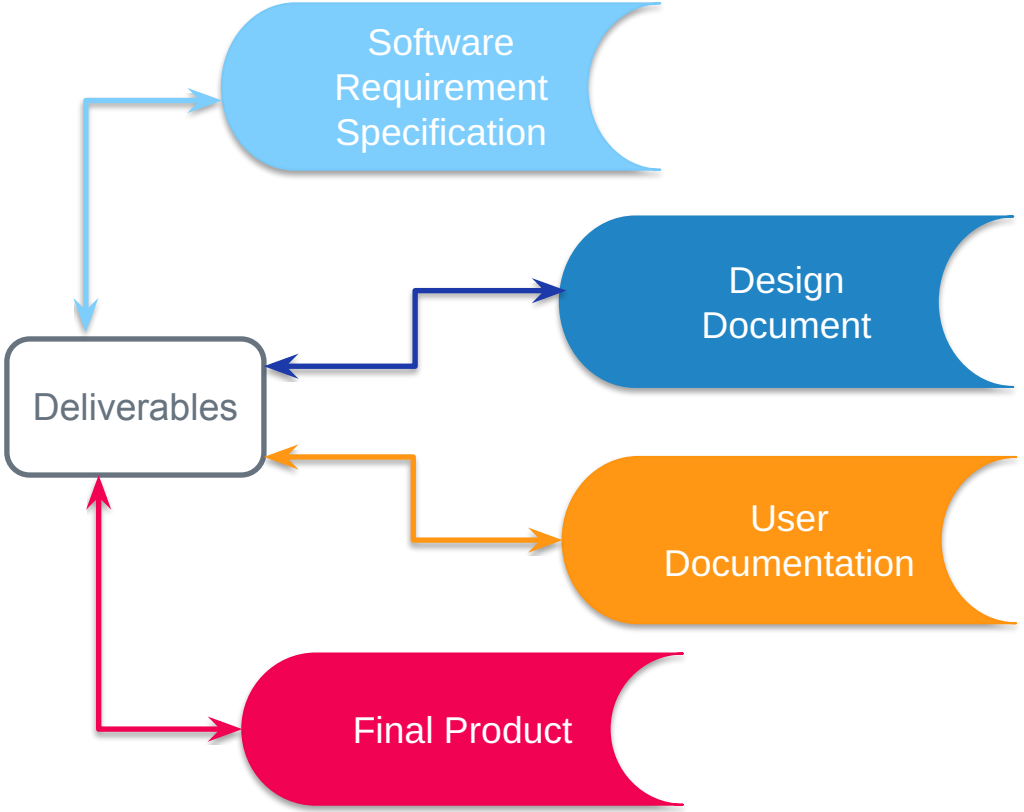
03.

To increase efficiency of processing documents.

04.

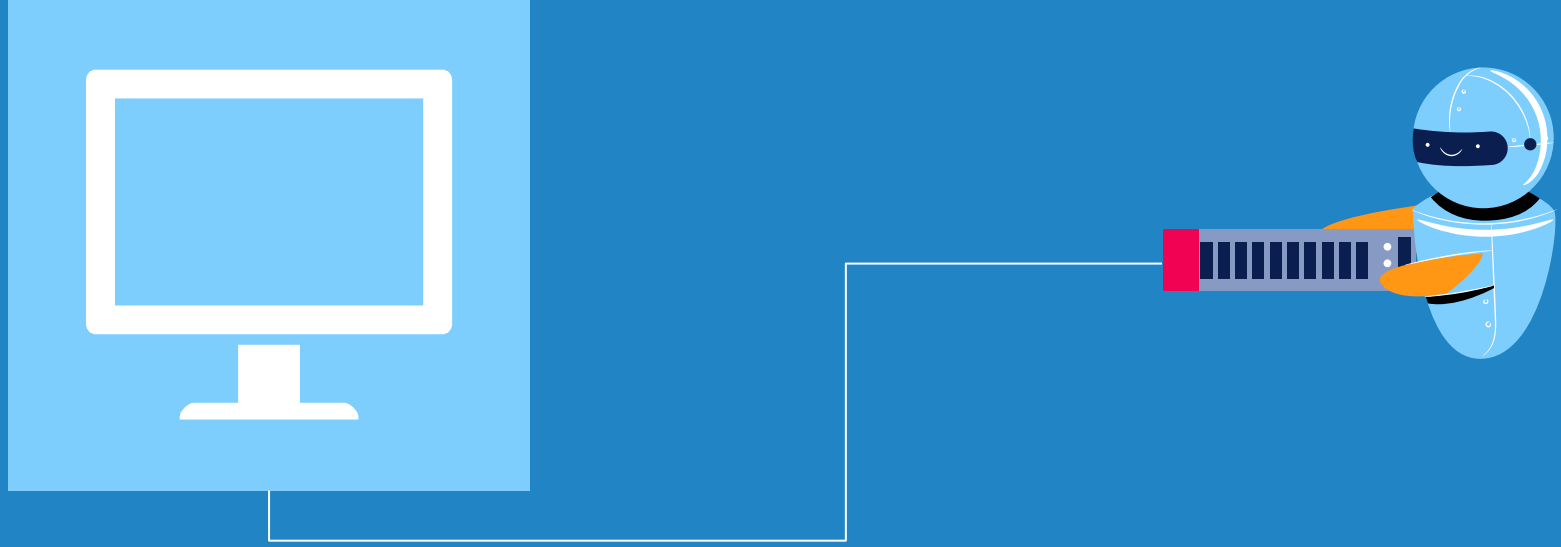
To minimize the resources required during manual process.

Deliverables



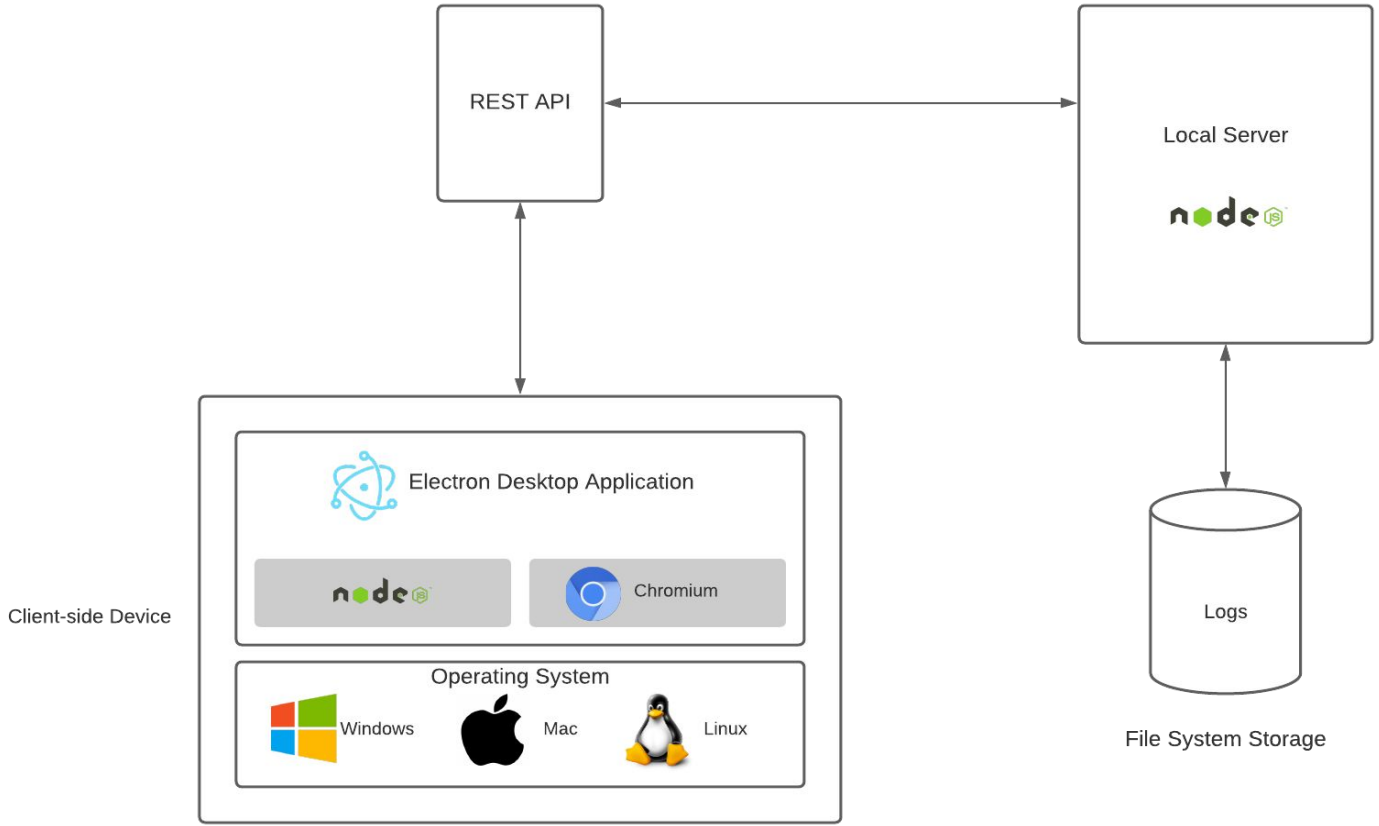
Literature Survey Conducted: State of Art

Title	Automated Extraction of Large Scale Scanned Document Images using Google Vision OCR in Apache Hadoop Environment. [(IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 9, No. 11, 2018]	Case Study on ABBYY Mobile OCR Engine for On-the-go loan approval used by DBS Bank, a leading financial services group in Asia.	onlineocr.net – Free online OCR service
Description	This paper proposes a text extraction method of document images using Google Vision OCR on the Hadoop architecture with twice the speed of manual data entry.	The proposed study is used for On-the-go loan approval by DBS Bank with the help of using OCR technology.	A Free web-based OCR service which uses REST cloud API to perform OCR on the input documents.

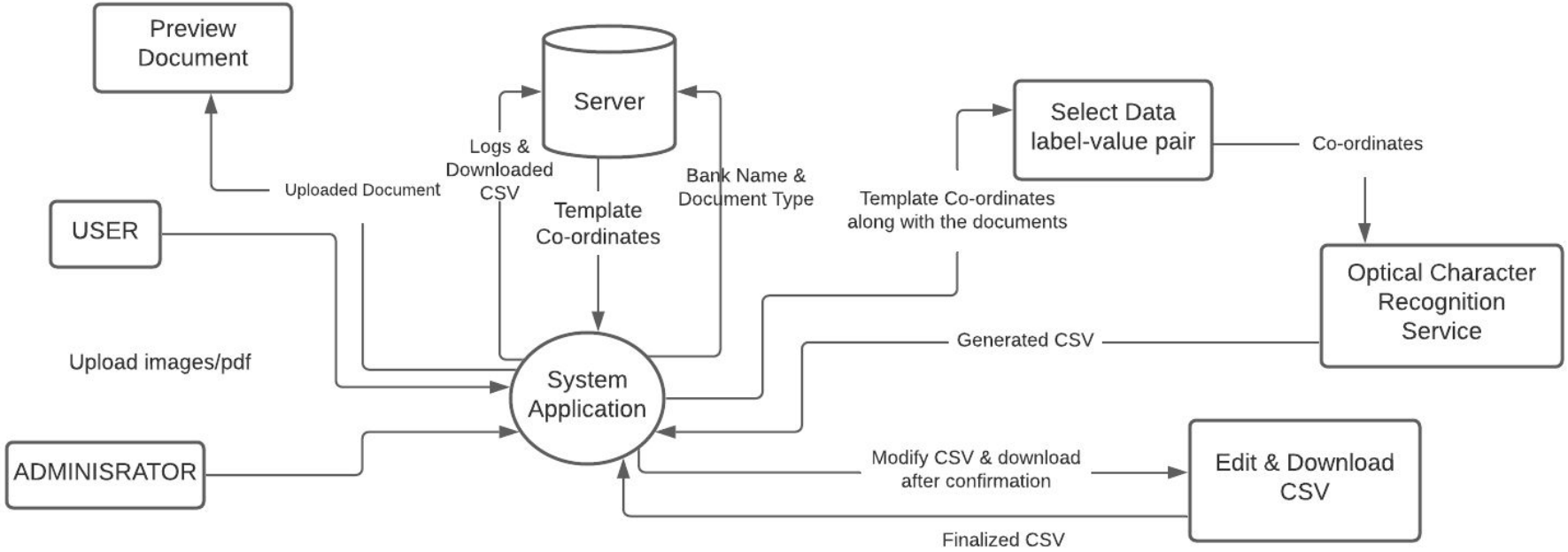


Project

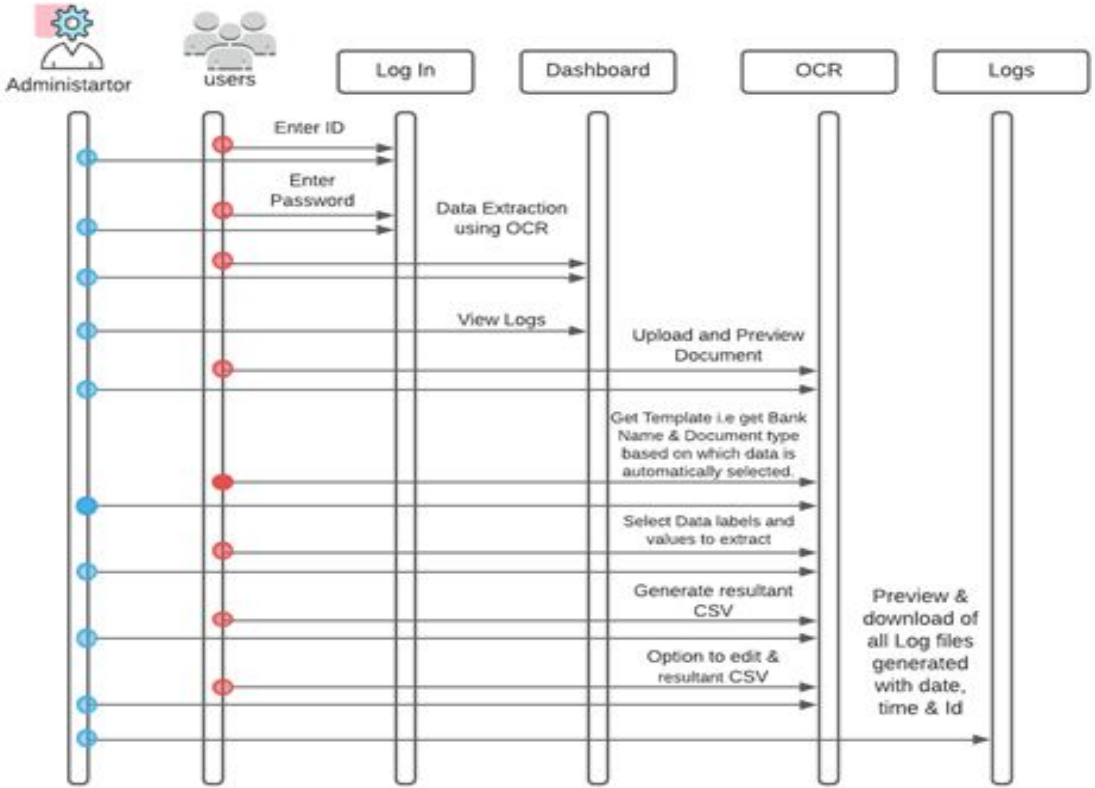
System Architecture

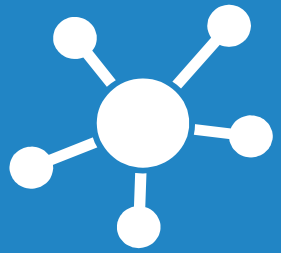


Data Flow Diagram



Sequence Diagram

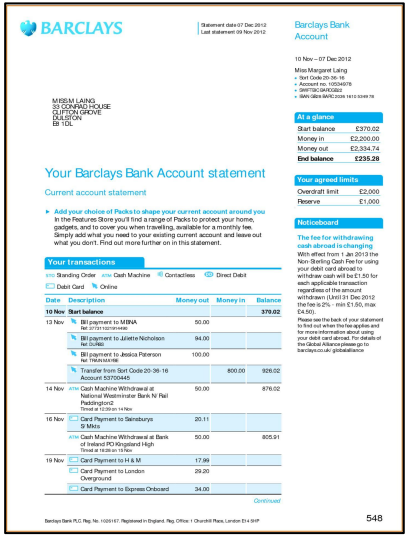




ALGORITHM

How it works ?

Step-wise Algorithm



BARCLAYS

Statement date 07 Dec 2012
Last statement 09 Nov 2012

MISS M LAING
33 CONRAD HOUSE
CLIFTON GROVE
DULSTON
E8 1DL

Barclays Bank Account

10 Nov – 07 Dec 2012

Miss Margaret Laing
- Sort Code 20-36-16
- Account no. 10534978
- IBAN GB84 BARC 2036 1610 3349 78

At a glance

Start balance	£370.02
Money in	£2,200.00
Money out	£2,334.74
End balance	£235.28

Your agreed limits

Overdraft limit	£2,000
Reserve	£1,000

Noticeboard

The fee for withdrawing cash abroad is changing

With effect from 1 Jan 2013 the Non-Sterling Cash Fee for using your debit card abroad to withdraw cash will be £1.50 for each applicable transaction regardless of the amount withdrawn (Until 31 Dec 2012 the fee is 2% - min £1.50, max £4.50).

Please see the back of your statement to find out when the fee applies and for more information about using your debit card abroad. For details of the Global Alliance please go to barclays.co.uk/globalalliance

Your Barclays Bank Account statement

Current account statement

► Add your choice of Packs to shape your current account around you
In the Features Store you'll find a range of Packs to protect your home, gadgets, and to cover you when travelling, available for a monthly fee. Simply add what you need to your existing current account and leave out what you don't. Find out more further on in this statement.

Your transactions

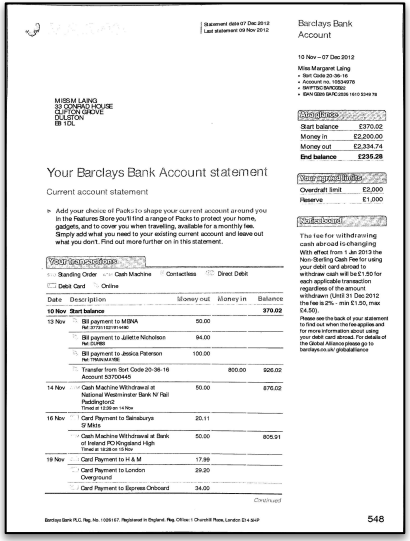
Standing Order · Cash Machine · Contactless · Direct Debit · Debit Card · Online

Date	Description	Money out	Money in	Balance
10 Nov	Start balance			370.02
13 Nov	Bill payment to MIMA Ref: 2013110219144E	50.00		
	Bill payment to Juliette Nicholson Ref: DJN89	94.00		
	Bill payment to Jessica Paterson Ref: TRAVAVAYRE	100.00		
	Transfer from Sort Code 20-36-16 Account 53700445		800.00	926.02
14 Nov	ATM Cash Machine Withdrawal at National Westminster Bank N Rail Paddington2 Timed at 12:09 on 14 Nov	50.00		876.02
16 Nov	Card Payment to Sainsburys S/ Mkts	20.11		
	ATM Cash Machine Withdrawal at Bank of Ireland PO Kingsland High Timed at 18:28 on 15 Nov	50.00		805.91
19 Nov	Card Payment to H & M	17.99		
	Card Payment to London Overground	29.20		
	Card Payment to Express Onboard	34.00		

Continued

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BARCLAYS

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DULSTON
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In the Features Store you'll find a range of Packs to protect your home, gadgets, and to cover you when travelling, available for a monthly fee. Simply add what you need to your existing current account and leave out what you don't. Find out more further on in this statement.

Your transactions

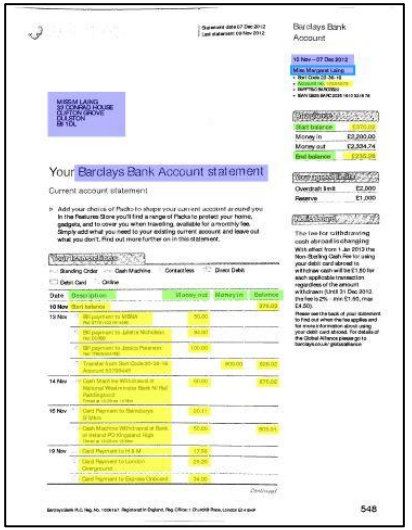
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19 Nov	Card Payment to H & M	17.99		
	Card Payment to London Overground	29.20		
	Card Payment to Express Onboard	34.00		

Continued

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Barclays Bank Account

Your Barclays Bank Account statement

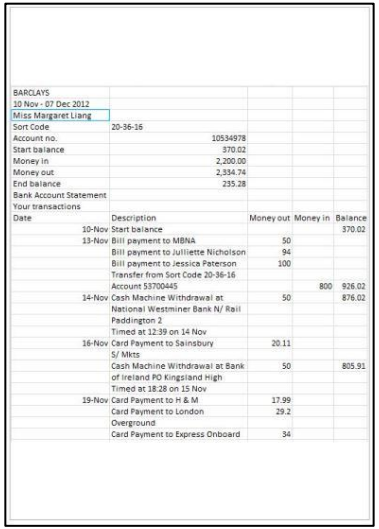
Current account statement

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Step-wise Algorithm

MISS M LAING
33 CONRAD HOUSE
CLIFTON GROVE
DULSTON
E8 1DL





BARCLAYS				
10 Nov - 07 Dec 2012				
Miss Margaret Liang				
Sort Code	20-36-16			
Account no.	10534978			
Start balance	370.02			
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Bank Account Statement				
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	Paddington 2			
	Timed at 12:39 on 14 Nov			
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	S/ Mkts			
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	of Ireland PO Kingsland High			
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	Card Payment to London	29.2		
	Overground			
	Card Payment to Express Onboard	34		

SPECIFICATIONS

Hardware

- LAN Connection
- RAM: 2GB
- Processor: 2.4 GHz
- Hard Disk: 1TB

Software

- Operating System: Windows, Linux
- Third Party API's: CSV File
- Commercially free Library: Tesseract OCR

Tech-Stack

























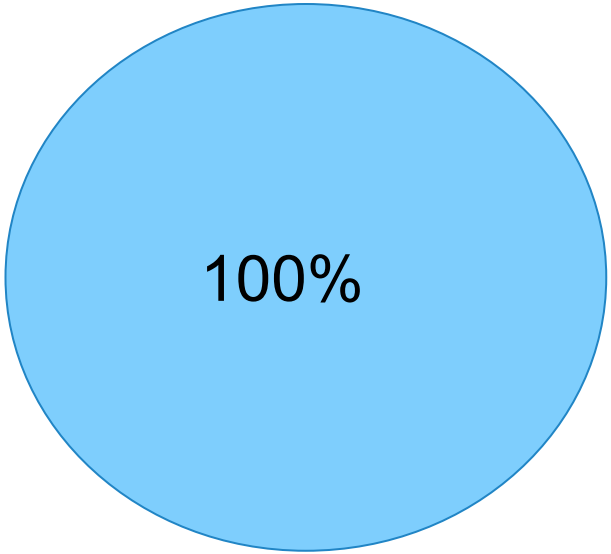
Tesseract OCR

System Implementation

Modules:

- ✔ Upload Document
- ✔ Preview Document
- ✔ Client Server Module
- ✔ Get Template Module
- ✔ Select Data Labels & values
- ✔ OCR Algorithm
- ✔ Edit CSV Module
- ✔ Download CSV Module
- ✔ Logs Module for Admin
- ✔ Login Module for Users and Admin





Deployment Strategy: The system is not the final product but a part of the main product, later to be integrated by company.

Testing: Some essential test cases

1. Test Case for Upload Document
2. Test Case for Get Template
3. Test Case for Select Data
4. Test Case for OCR Working
5. Test Case for Edit & Download CSV

1. UPLOAD DOCUMENT

Sr. No	Test	Steps to Execute	Test i/p	Test o/p	Status
1.	User Interface	Click on the buttons & other components for proper functioning	NULL	Proper UI Functioning	PASS
2.	Browse & Select Digital Image File	Click on the Browse Button Select the images in jpeg, jpg etc. image formats Click ok After Selected	Documents should be of jpeg, jpg, png etc digital image formats only	Documents should be successfully uploaded	PASS

Continued .....

1. UPLOAD DOCUMENT

3.	Check Exceptions for file size.	Click on the Browse Button Select the documents in image formats where of size more than 15 MB Click ok After Selected	Documents should be of jpeg, jpg, png etc digital image formats only of size more than 15 MB	Should raise an exception and document will not be uploaded.	PASS
4	Check Exceptions for file type.(For eg: mp4 file)	Click on the Browse Button Select an mp4 file. Click ok After Selected	Documents should be of mp4 format only	Should raise an exception and document will not be uploaded.	PASS

2. GET TEMPLATE

Sr. No	Test	Steps to Execute	Test i/p	Test o/p	Status
1.	User Interface	Click on the buttons & other components for proper functioning	NULL	Proper UI Functioning	PASS
2.	Choose Bank Name	Click on the Bank Name Dropdown Box. Select the required Bank.	Select Bank Name.	Bank Name should get selected.	PASS
3	Choose Document Type.	Click on the Document Type Dropdown Box. Select the Required Document Type.	Select Document Type.	Document Type should get selected.	PASS

3. SELECTING DATA LABELS AND VALUES

Sr. No	Test	Steps to Execute	Test i/p	Test o/p	Status
1	Choose a different doc from different bank for extraction of data.	Click on the back tooltip button provided on the bottom left.	NULL	Should redirect back to the get template page.	PASS
2	Data Marked based on Bank name & Document type.	After get template functioning, click on next.	Bank name & Document type.	Data should be automatically selected based on Bank Name & document type.	PASS

Continued .....

3. SELECTING DATA LABELS AND VALUES

3	Manually select data labels & values	Draw a rectangle on the data label or value you wish to select apart from the data automatically selected. Select the check button on the left hand side.	Document	Successful selection of data labels and values.	PASS
4	Deselect data	Draw a rectangle on data label or value you wish to deselect. Click on the cross button provided on the left hand side.	Document	Data should get deselected.	PASS

4. WORKING OF OCR

Sr. No	Test	Steps to Execute	Test i/p	Test o/p	Status
1	Download CSV	Make required changes in editable CSV generated. Give a final Check. Click on the download tooltip button on the bottom right for downloading.	Generation of Editable CSV	CSV should be downloaded on the device.	PASS
2.	Data Extraction	After selection of data labels & values, click on the next tooltip button on the bottom right.	Data must be selected.	Selected Data is generated in target CSV File.	PASS

5. EDIT AND DOWNLOAD CSV

Sr. No	Test	Steps to Execute	Test i/p	Test o/p
1.	Download CSV	Make required changes in editable CSV generated. Give a final Check Click on the download tooltip button on the bottom right for downloading.	Generation of Editable CSV	CSV should be downloaded on the device.
2	Editable CSV Generation	After successful data selection, click on next. Click of the Editable CSV generated along with the actual uploaded document. Make required changes.	NULL	Editable CSV should be generated.

Future Scope



As banking continues to transform with each passing day, the proposed system too has a few aspects to implement in respect to the future work

- Extracting data from a handwritten document
- Support Multilingual document

Paper Publication



Review Paper under the name “ A Detailed Review on Text Extraction using Optical Character Recognition” has been accepted for publication by SPRINGER PUBLICATIONS & will be published soon.

Conclusion

- Working with clients in real life
- Planning and Prototyping gives clear vision
- Consideration for Adaptability and Integrability
- Understood OCR algorithm
- Choosing the right tools
- Deployment with security
- Adapting to the online world

References

[1] https://towardsdatascience.com/what-is-ocr-7d46dc419eb9

[2] R. Smith, “An Overview of the Tesseract OCR Engine,” in Proc. International Conference on Document Analysis and Recognition, 2007.

[3]https://www.researchgate.net/publication/331948679_Automated_extraction_of_large_scale_scanned_document_images_using_Google_Vision_OCR_in_Apache_Hadoop_en Vironment

[4]https://levelup.gitconnected.com/text-extraction-from-a-table-image-using-pytesseract-and-opencv-3342870691ae

Thank you!

Feel free to ask any questions :)